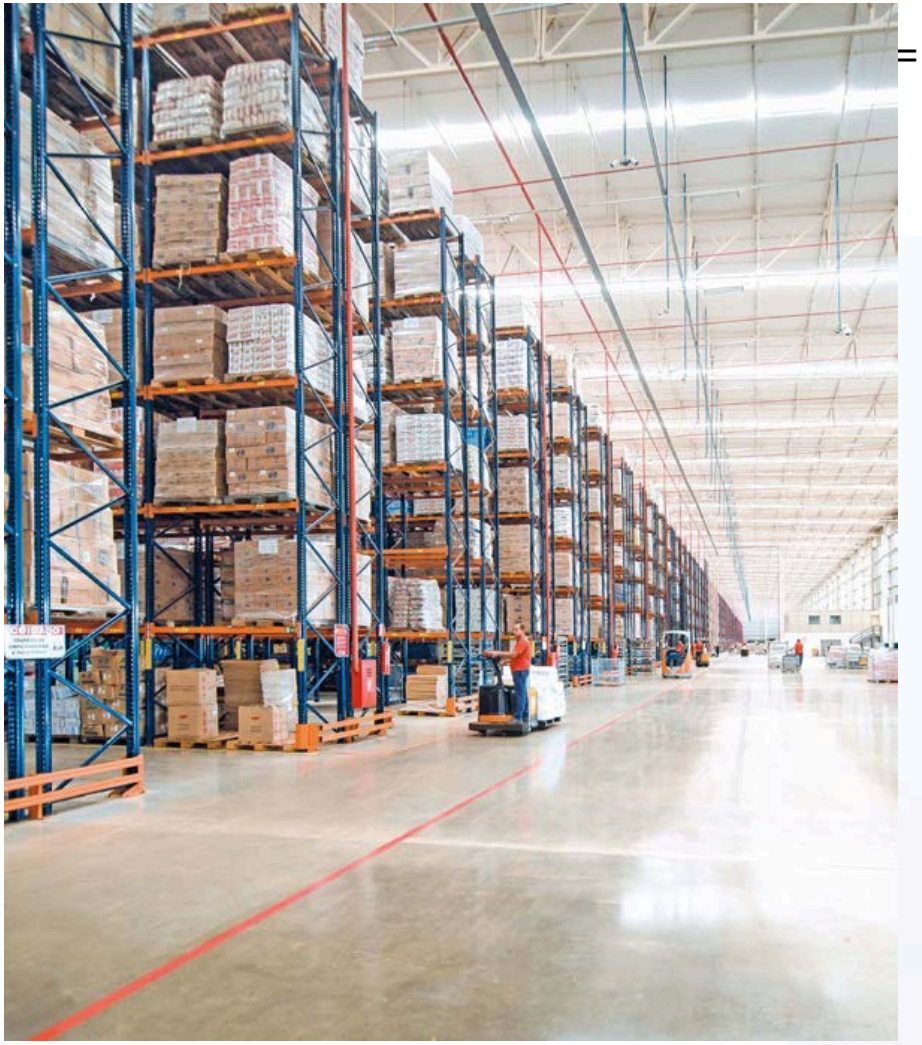
MECALUX

MECALUX

HACKUPC 2026:
WAREHOUSE OPTIMIZER

ANATOMY OF A RACK

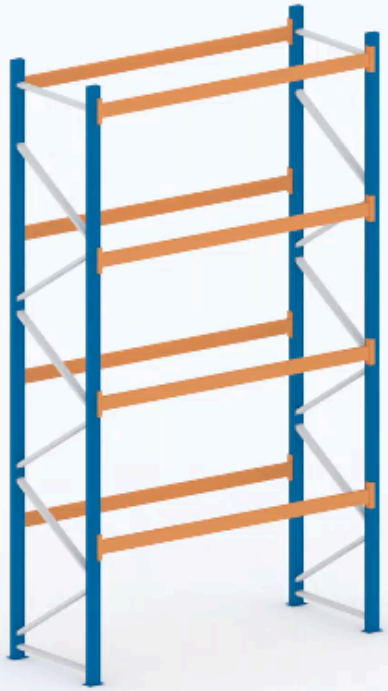


=

Frames + Beam =

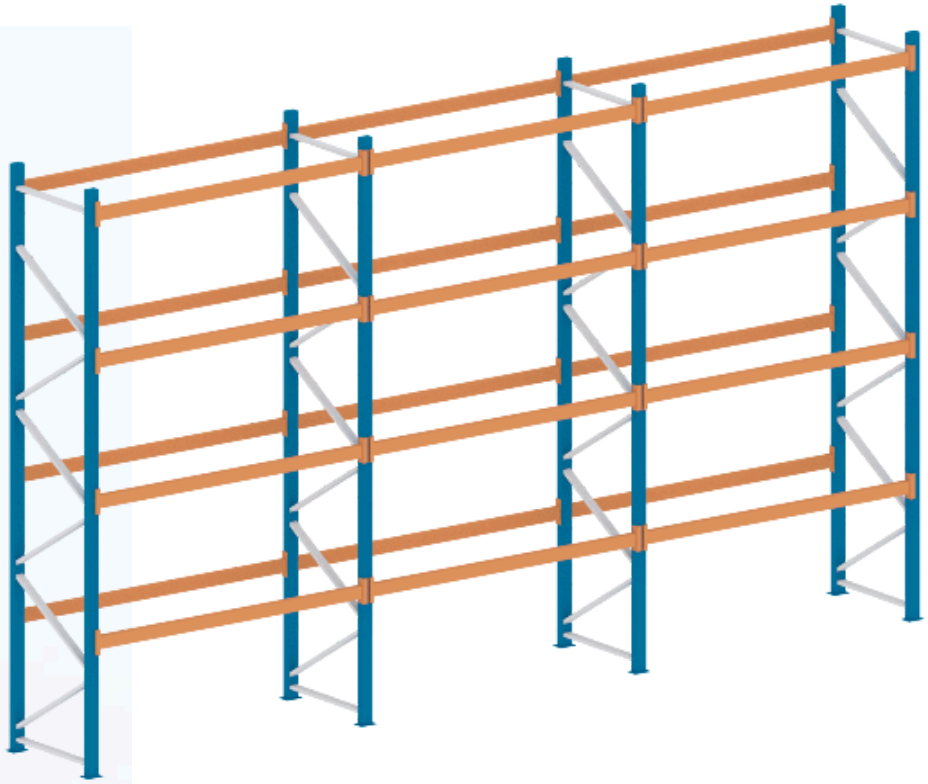
Bay

:



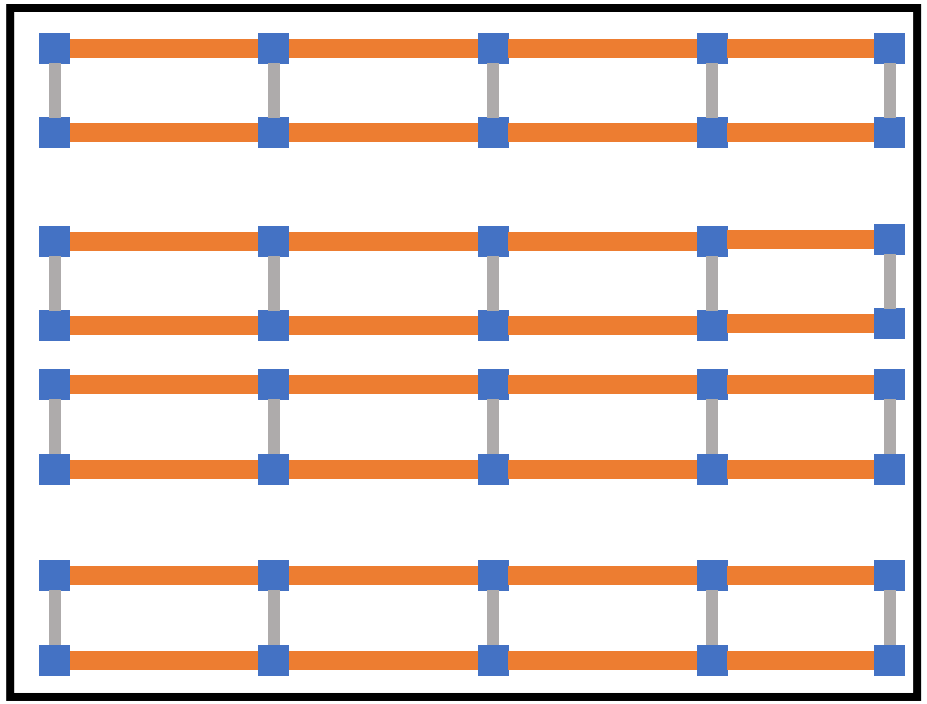
Many Bays =

Row:

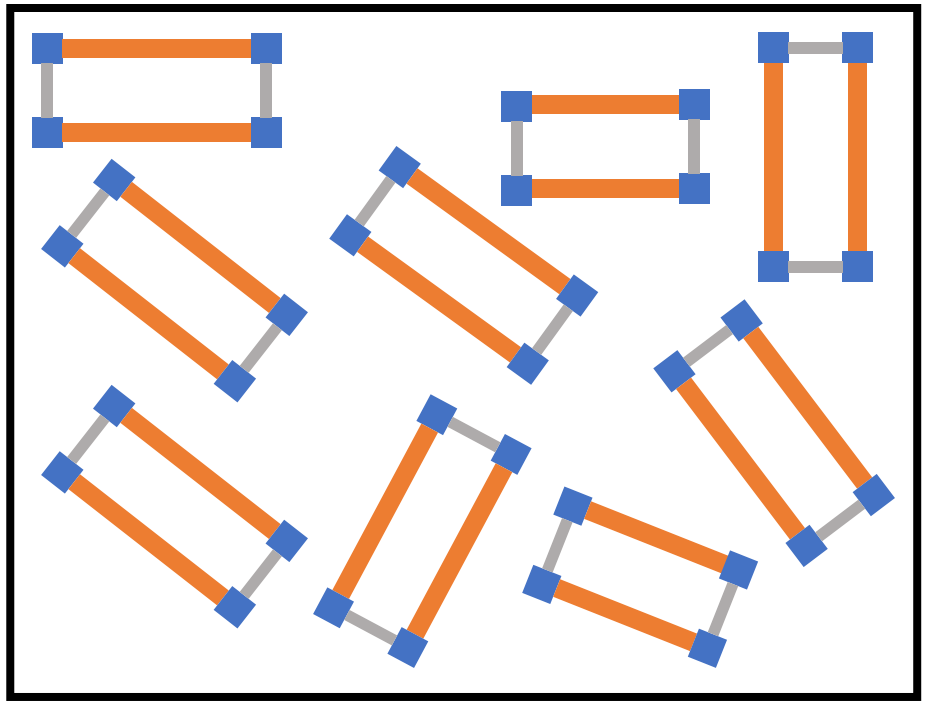


MECALUX

OUR PROBLEM



or



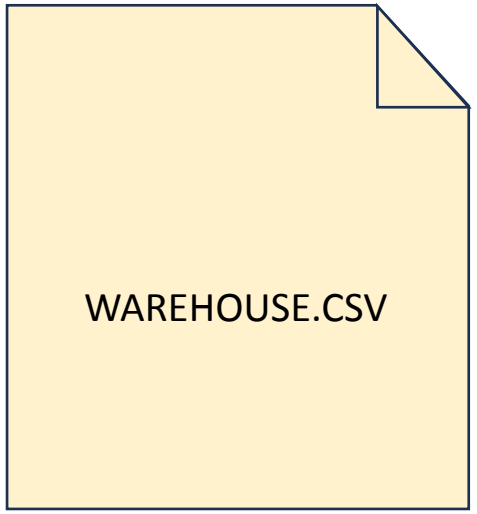
?

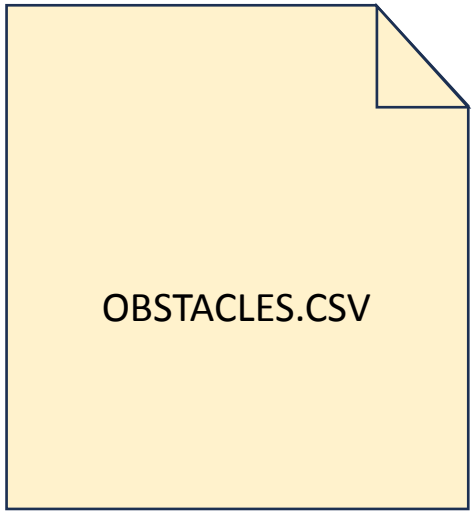
Your goal: **Place bays** in a warehouse the **cheapest way** using the largest ammount of area

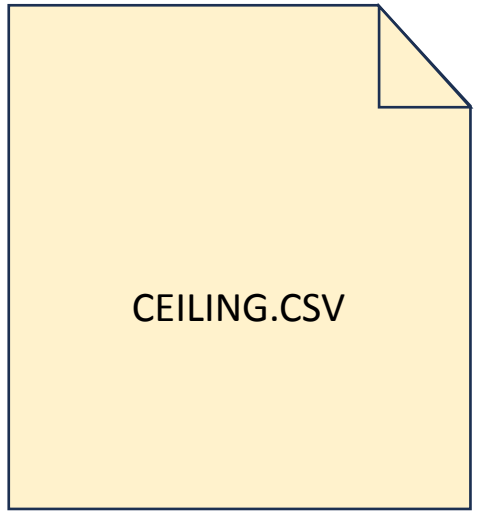
MECALUX

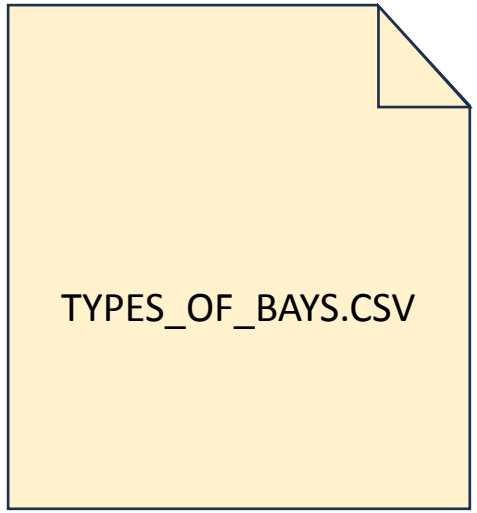
INPUT

Each case consists on 4 CSV files





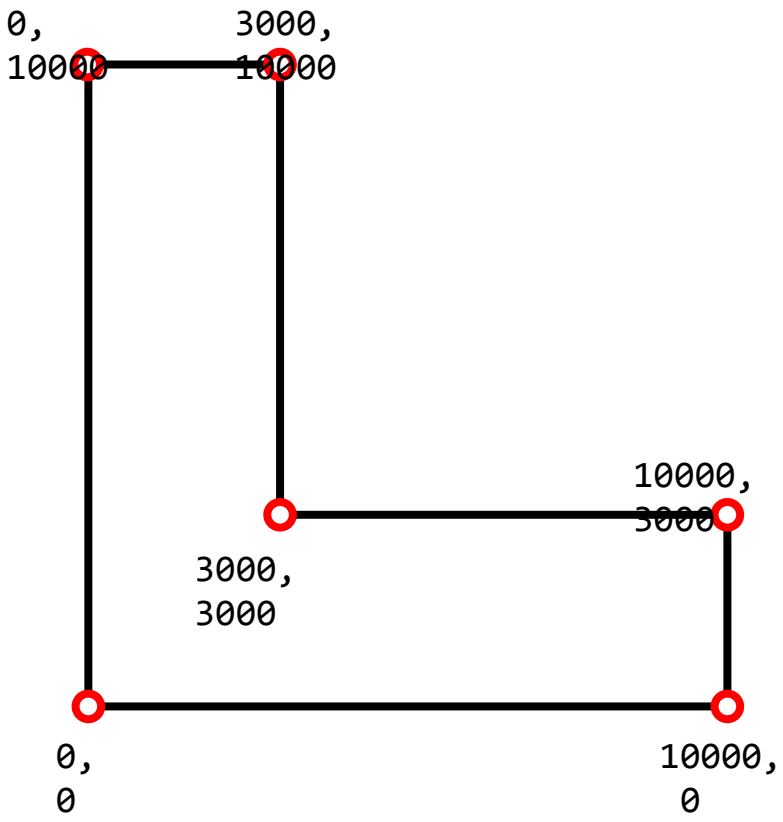




INPUT: WAREHOUSE

Coord X, Coord Y

```
0, 0
10000, 0
10000, 3000
3000, 3000
3000, 10000
0, 10000
```



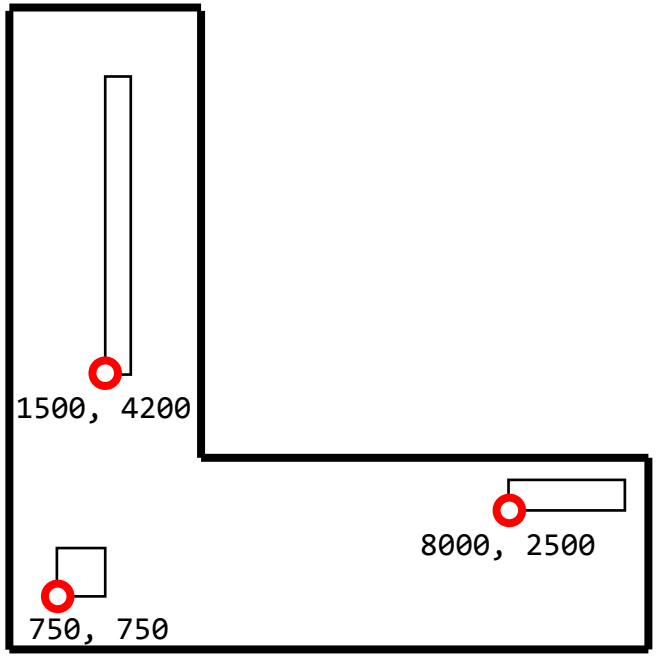
The walls of the warehouse will always be axis aligned

MECALUX

INPUT: OBSTACLES

Coord X, Coord Y, Width, Depth

```
750, 750, 750, 750
8000, 2500, 1500,
300
1500, 4200, 200, 4600
```

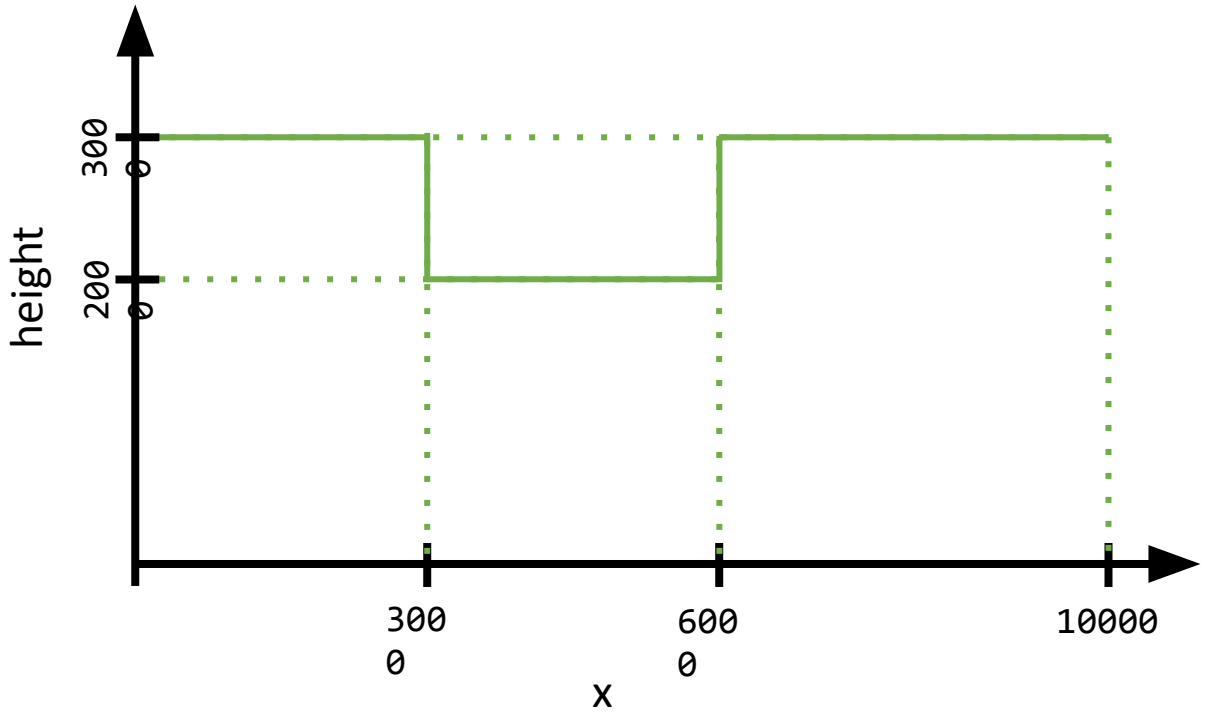


The obstacles will always be boxes

INPUT: CEILING

Coord X, Ceiling Height

```
0, 3000
3000, 2000
6000, 3000
```

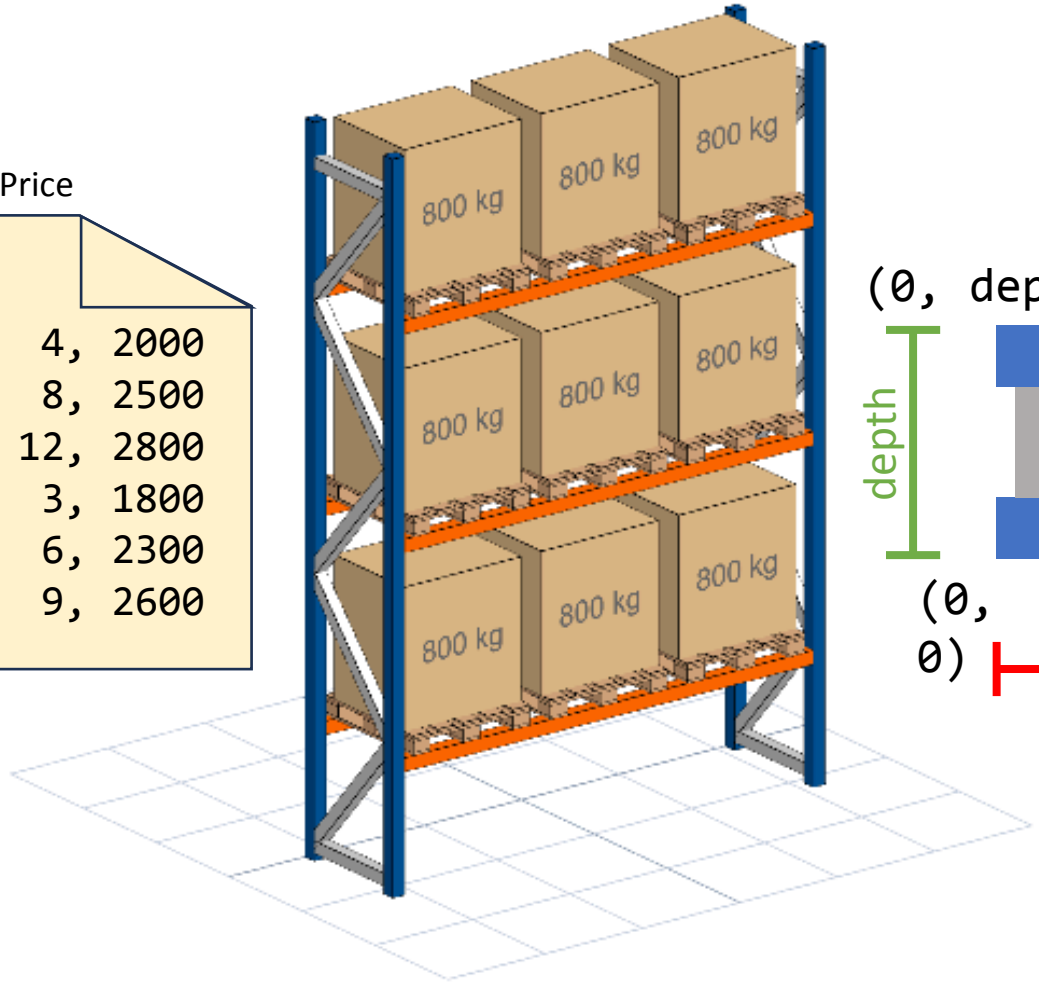


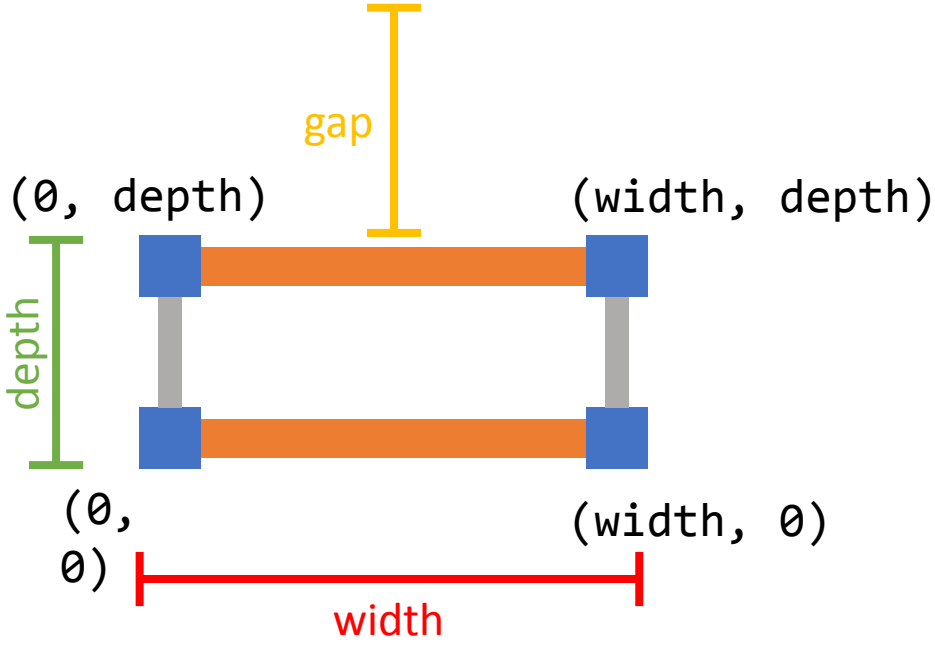
MECALUX

INPUT: BAY

Id, Width, Depth, Height, Gap, nLoads, Price

```
0,  800, 1200, 2800, 200,  4, 2000
1, 1600, 1200, 2800, 200,  8, 2500
2, 2400, 1200, 2800, 200, 12, 2800
3,  800, 1000, 1800, 150,  3, 1800
4, 1600, 1000, 1800, 150,  6, 2300
5, 2400, 1000, 1800, 150,  9, 2600
```



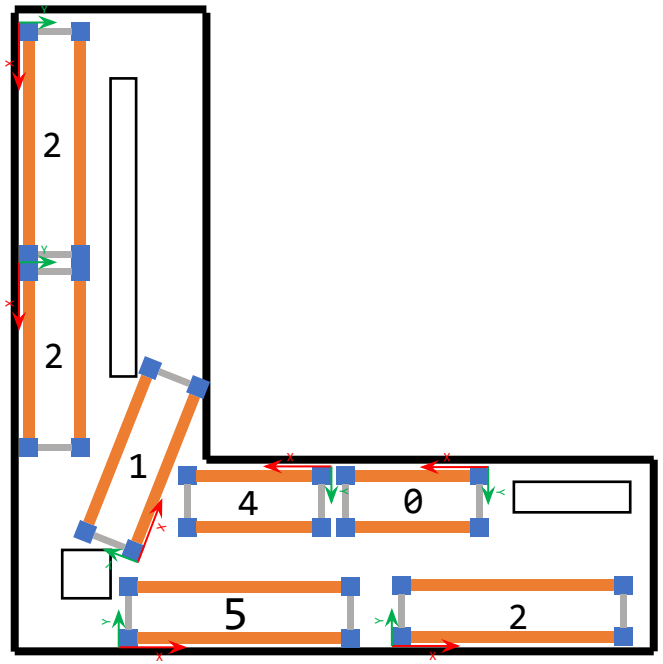


MECALUX

EXPECETED OUTPUT

Id, X, Y, Rotation

```
0, 6600, 2900, 180
1, 1000, 1600,  70
2,    0, 5400, 270
2,    0, 9900, 270
2, 5000,    0,   0
4, 4400, 2900, 180
5, 2000,    0,   0
```



MECALUX

EVALUATION

Solution Generation

- Number of correct solutions
- Quality of the solution:

$$Q = \left(\sum_{bay} \frac{price}{loads}\right)^{2-\frac{\sum_{bay} area}{area_{warehouse}}}$$

Presentation

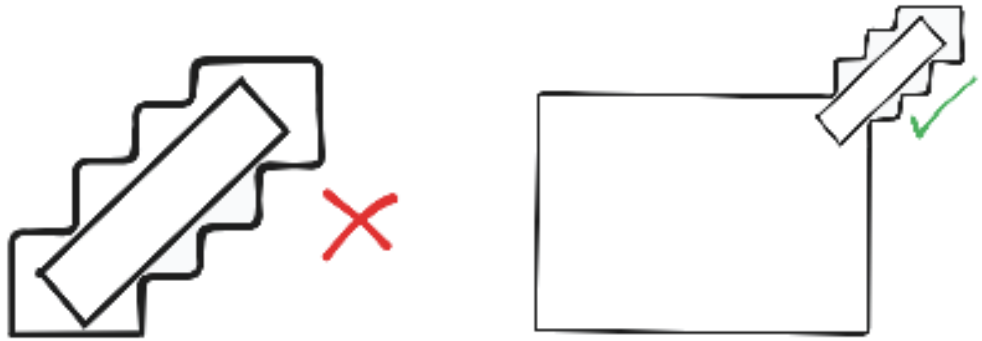
- Visualization of the solution
- Whatever UI features

MECALUX

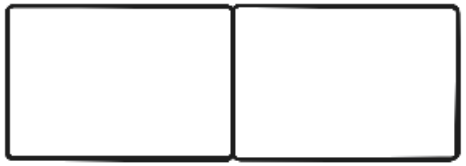
JUDGING

- During judging you will need to run some tests on your machine
  - Make sure it does not take more than 30s to finish

- Judging tests won't be crazy edge cases.
  - They may be bigger than the examples



- Bays can share boundaries, same with warehouse
  - If a bay at position (0,0) has width 1000, you don't need to put the next one at (1001,0), (1000,0) is fine.



ANY QUESTION?

GOOD HACKING

MECALUX